Assignment
CCS23213 Data Mining
Real-World Data Mining Project

Objective: The goal of this assignment is for students to apply data mining techniques to a real-world dataset, generate insights, and analyse the results. Students are required to identify a suitable problem, select an appropriate dataset, perform data pre-processing and exploratory data analysis, build data mining models, evaluate model performance, and provide meaningful conclusions and recommendations. This assignment carries 20% of your grade.

Submission date: Week 13 (18 June, 2026) via Moodle
Presentation date: Week 14 (24/25 Jun 2026)

Mode of assignment: Group of 3

Group and Dataset Registration: https://bit.ly/CCS2313-Group
(*Note: Ensure your group registration and dataset choice are confirmed before proceeding*)

Instructions:

1. Problem Identification & Dataset Selection

Choose a dataset from a public source such as Kaggle, UCI Machine Learning Repository, government data portals, or other reliable open data sources. The dataset should have at least 400 records and 5 attributes, and must be suitable for a classification or prediction problem.

Describe the dataset in detail including:
- The background of the dataset, source, number of attributes, and number of records.
- The type of each attribute according to its measurement scale, namely nominal, ordinal, interval, or ratio.
- Clearly define the problem that you intend to solve.
- State the objective of the analysis.
- Identify the type of data mining task involved, either classification or prediction.
- Determine the two data mining algorithms that you plan to use.
- Justify the choice of algorithms based on the problem and dataset.

2. Data Preprocessing

Perform suitable data pre-processing on the dataset. This may include:
- handling missing values;
- detecting and handling outliers;
- checking data distribution; and
- encoding categorical variables, if required;
- applying normalization, if required; and
- applying feature selection, if required.

For each pre-processing step, explain:
- the problem identified;
- the method used;
- the reason for choosing the method; and
- the result after pre-processing.

Students are required to use Exploratory Data Analysis (EDA), including summary statistics and visualizations such as histograms, scatterplots, box plots, and correlation matrices. Students must also detect skewness and outliers and provide appropriate interpretation.

Report the methods used to identify data problems and explain how these problems were resolved. Provide justification for all techniques applied. Explain the insights derived from the

EDA and how they inform further analysis. Relevant graphs and charts must be included in the report to support the findings.

3. Model Building

- Choose and implement two data mining algorithms for either classification or prediction.
- Evaluate model performance using appropriate metrics. Perform train-test split experiments using the following five splits: 50:50, 60:40, 70:30, 80:20, and 90:10.
- For classification, use suitable metrics such as Accuracy, Precision, Recall, and F1-score.
- For prediction, use suitable metrics such as MAE, MSE, RMSE, and R-squared.
- Show a suitable heatmap, such as a correlation heatmap or confusion matrix heatmap, where applicable.
- Identify the best train-test split for each algorithm and determine the best overall model. Justify your decision.
- Validate the selected best model using five examples of unseen input and present the predicted output.
- Summarize your findings and provide actionable recommendations based on the data mining results. Discuss the strengths and limitations of your analysis.

Example of Model Evaluation Table

a. Classification

No	Fold Split	Algorithm 1				Algorithm 2			
		Accuracy	Precision	Recall	F1-Score	Accuracy	Precision	Recall	F1-Score
1	50:50								
2	60:40								
Best Fold By Metrics									

Best Model (Overall):
Insight Analysis:

Heatmap image

b. Prediction

No	Fold Split	Algorithm 1				Algorithm 2			
		MAE	MSE	RMSE	R^2	MAE	MSE	RMSE	R^2
1	50:50								
2	60:40								
Best Fold By Metrics									

Best Model:
Insight Analysis:
Heatmap image

* Ensure that your report is well-organized, clear, and follows a logical structure: Introduction, Methods, Results, Discussion, Conclusion, References, and Appendix if necessary.